

IT5117 · PROGRAMMING ESSENTIALS FOR AI

Daily Rainfall Estimation & Weather Analysis

for Anuradhapura, Sri Lanka

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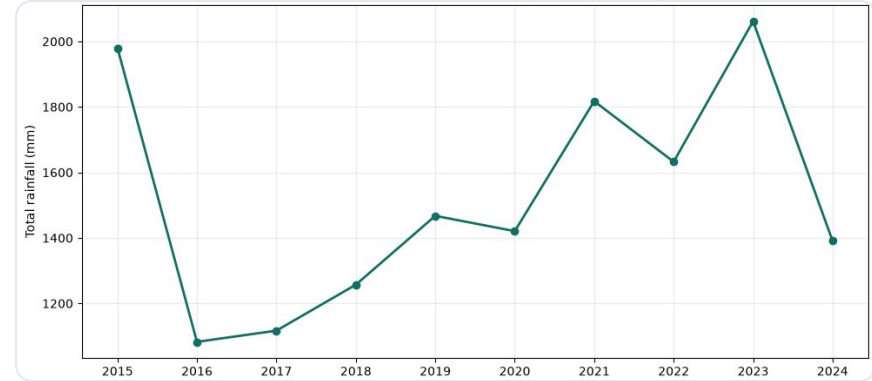
Rainfall variability makes agricultural planning difficult.

Decision Context

- Seasonal rainfall is uneven, and extreme events remain difficult to predict.
- Farmers and planners need local evidence to make informed decisions.

Agricultural Importance

- Seasonal rainfall directly affects crop timing, irrigation scheduling, water storage capacity, and weather preparation.
- The primary analysis objective is to estimate daily rainfall and identify key local patterns.
- Annual data demonstrates substantial year-to-year variation, motivating the need for consistent historical analysis.



Ten Years of Data Show a Clear Rainfall Season

NASA POWER Agroclimatology Parameters

- **Dataset:** 3,653 complete daily records (no missing -999 values) sourced from NASA POWER.
- **Scope:** Covers 2015 to 2024, including 15 weather measurement variables with no personal data.
- **Target Variable:** PRECTOTCORR (mm/day) utilized for estimation.
- **Seasonality:** Monthly analysis reveals a clear seasonal pattern, with November experiencing the highest average daily rainfall at ~11.45 mm.

Key Meteorological Variables Analyzed

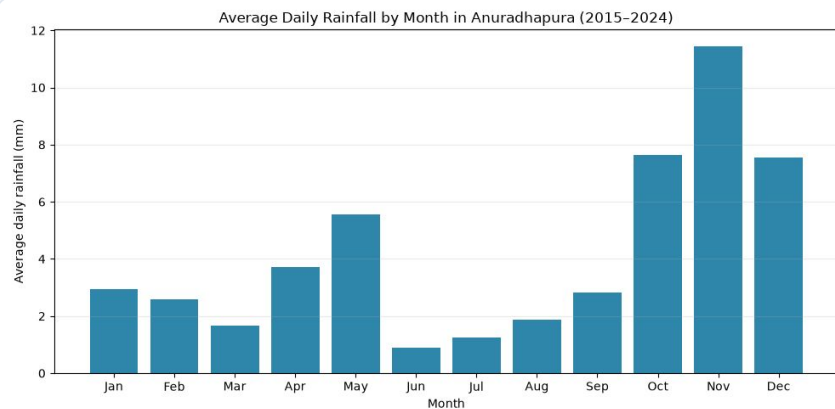
Pressure

Specific Humidity

Cloud Amount

Wind Direction

Dew Point



Two-Stage Modelling

Separates Rainfall Occurrence from Rainfall Amount

01

01 / PREPARATION

Prepare Trustworthy Data

The workflow begins by validating the 15 variables. It creates calendar, lag, and rolling-rainfall features utilizing past days only to ensure leakage-safe features.

The output is a clean 3,653-row table. Training covers the period from 2015 to 2022.

02

02 / EVALUATION

Primary Regression

The main solution compares Linear Regression, Random Forest, and Gradient Boosting models on the chronological 2023–2024 test set.

The output provides rainfall estimates and residuals for error and variance analysis.

03

03 / TWO-STAGE EXTENSION

Rainfall Occurrence & Amount

The extension separates the question of whether it rains from the amount. It first classifies rain of ≥ 1 mm. If detected, it estimates the amount using conditional regression.

The output includes a validated confusion matrix and the estimated rainfall amount.

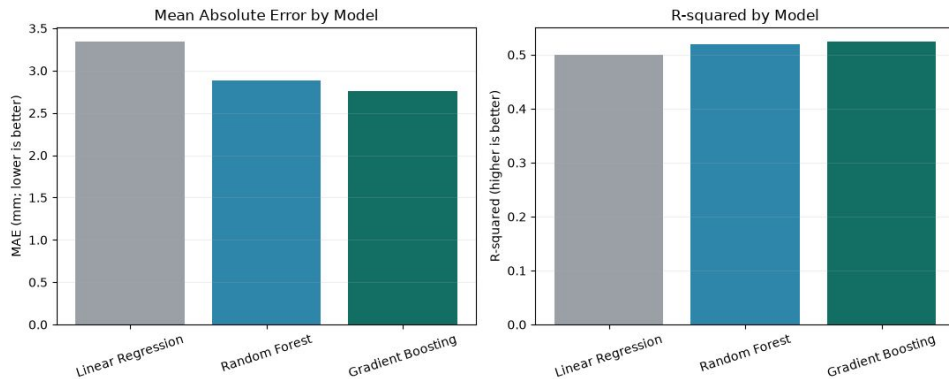
Gradient Boosting is the strongest main model.

Chronological Evaluation

- Achieved a mean absolute error (MAE) of 2.756 mm on unseen 2023–2024 data.
- Root Mean Squared Error (RMSE) is 6.602 mm, and the R^2 value is 0.524.

Why it remains the main model

- Demonstrated the best MAE among the three evaluated regressors.
- Possesses the appropriate complexity for the module.
- Evaluates rain occurrence separately and estimates amount via conditional regression on rainy days.



MEAN ABSOLUTE ERROR (MAE)

2.756 mm

ROOT MEAN SQUARED ERROR (RMSE)

6.602 mm

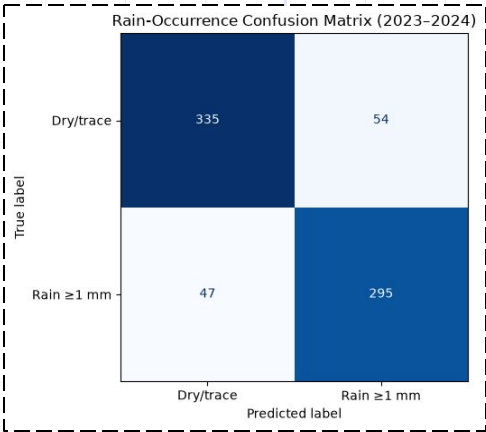
COEFFICIENT OF DETERMINATION (R^2)

0.524

Rainfall Detection & Estimation Performance

Evaluation of the two-stage model over 731 test days reveals **86.2% classification accuracy**

ACTUAL CLASS	PREDICTED CLASS	
	DRY / TRACE	RAINY
	TRUE NEGATIVE 335 Correct Dry/Trace Days	FALSE POSITIVE 54 False Rain Alerts
	FALSE NEGATIVE 47 Missed Rainy Days	TRUE POSITIVE 295 Correct Rainy Days



Rain Classification

PRECISION

0.845

Positive predictive value

RECALL

0.863

Sensitivity of detection

F1-SCORE

0.854

Harmonic mean measure

Amount Estimation

MEAN ABSOLUTE ERROR

2.754 mm

Average deviation amount

R² COEFFICIENT

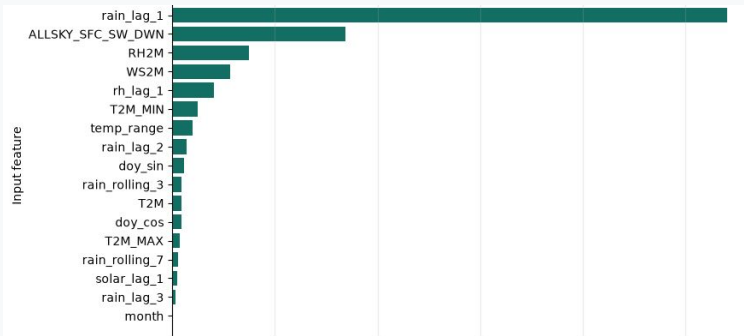
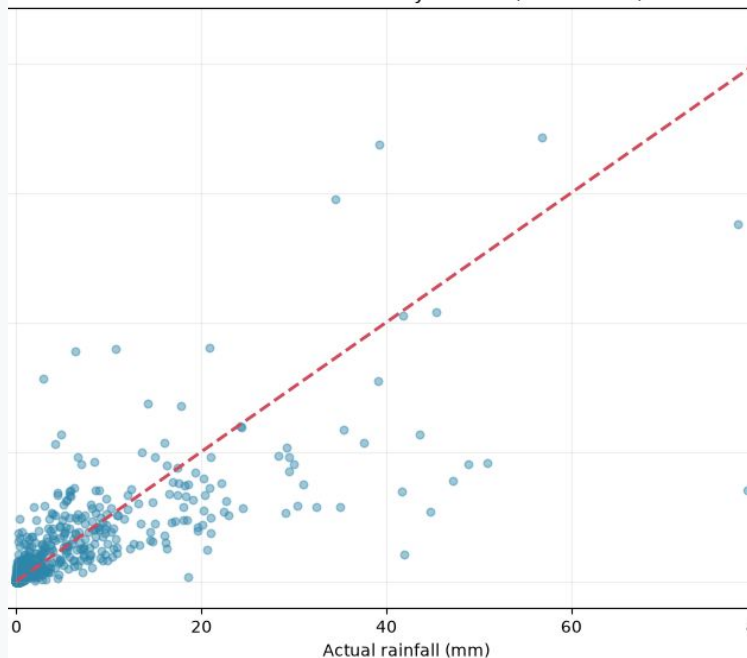
0.530

Goodness-of-fit metric

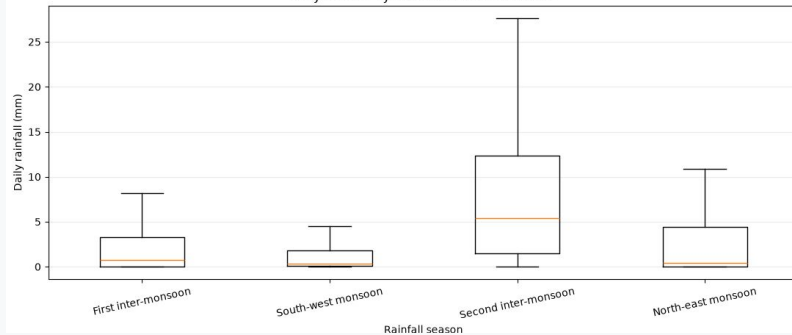
Regression metrics are conditional on true positive occurrences.

Additional Charts Explain Model Behaviour

Actual vs Estimated Daily Rainfall (2023-2024)



Daily Rainfall by Sri Lankan Rainfall Season



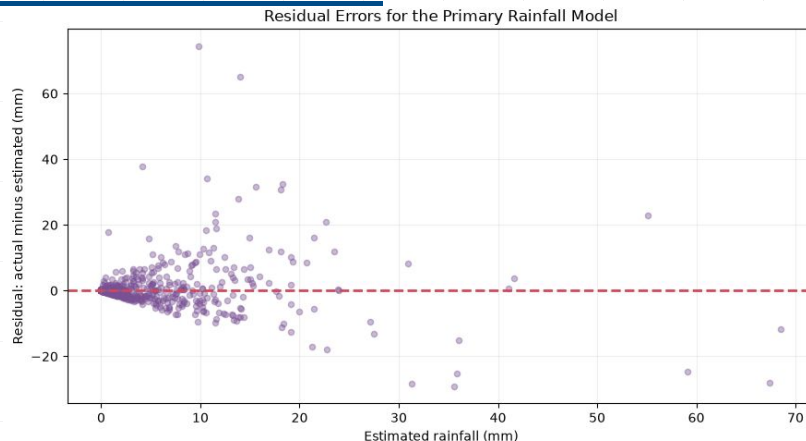
Extreme rainfall remains the model's main limitation.

What the errors show:

- Residuals concentrate near zero on low-rainfall days.
- Rare heavy events create much larger errors, showing difficulty with extremes.
- Functions as an estimator rather than an operational forecast.

Future Improvements:

- Utilize verified rainfall measurements from local weather stations.
- Integrate real weather forecast data for genuine predictive capability.



Practical Conclusion:

The project is highly useful for historical analysis and current-day estimation. **A key limitation is that NASA POWER provides gridded data, not direct station data, meaning heavy rainfall needs further work.**

Data Sources & Acknowledgements

Data Source & Documentation

[1] NASA POWER Project Daily API documentation
power.larc.nasa.gov/docs/services/api/temporal/daily/

[2] NASA POWER Project Parameter dictionary
power.larc.nasa.gov/parameters/

Data Request & Context

[3] NASA POWER dataset request (Anuradhapura · 2015–2024 · AG community)
power.larc.nasa.gov/api/...format=CSV

[4] Department of Census and Statistics, Sri Lanka (Area and climate information)
statistics.gov.lk/abstract2024/chapter1

Model Documentation

[5] Gradient Boosting Regressor (scikit-learn)
scikit-learn.org/stable/...GradientBoostingRegressor.html

[6] Random Forest Regressor (scikit-learn)
scikit-learn.org/stable/...RandomForestRegressor.html

[7] Random Forest Classifier (scikit-learn)
scikit-learn.org/stable/...RandomForestClassifier.html

Acknowledgement

The dataset was downloaded from NASA POWER.



Thank you

